LDT527: Designing Constructivist Learning Environments

The information contained on this page is designed to give students a representative example of material covered in the course. Any information related to course assignments, dates, or course materials is illustrative only. For a definitive list of materials, please check the online catalog 3-4 weeks before the course start date.

LDT 527 Designing Constructivist Learning Environments (3 credits): Designing learning environments based on constructivist principles of learning that provide modeling, coaching, and scaffolding.

Course Prerequisites: Recommended (not required): LDT 415; prior learning theory class.

Overview | Objectives | Materials | Library Resources | Technical Requirements and Help | Course Requirements and Grading | Course Schedule | Academic Integrity | Student Success and Support | Policies

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Overview

This course emphasizes the design of learning environments based on constructivist principles of teaching and learning. The role of technology in the design of learning environments has become a topic of great attention in educational research, technology, and practice. This is an active area of educational research, and, at this time, research strategies and analytical techniques are still evolving and expanding. The increasing number of design efforts and technology advances call for establishing more formalized approaches to designing learning environments. Course participants will work together to develop a better understanding of the principles involved in student-centered, constructivist learning. Participants will create “blueprint” designs of a variety of learning environments that are grounded in research and/or emerging theory of student-centered, constructivist learning.

Course Objectives

Upon completion of this course, you should be able to

- develop a better understanding of the principles involved in student-centered, constructivist learning; and
- create “blueprint” designs of a variety of learning environments that are grounded in research and/or emerging theory of student-centered, constructivist learning.

Required Course Materials

No textbooks or other materials to purchase. Readings for the course are available on electronic reserve (E-Reserves) from the University Libraries (via Library Resources on the Course Navigation Menu) and/or as links from the Course Schedule.

Most World Campus courses require that students purchase materials (e.g., textbooks, specific software, etc.). To learn about ordering materials, please see the Course Materials page (http://student.worldcampus.psu.edu/course-work-and-success/course-materials-and-library-services/course-materials) . Before the course begins, you should check LionPATH for a list of required materials.

Library Resources

This course requires that you access Penn State library materials specifically reserved for this course. You can access these materials by selecting Library Resources in your course navigation, or by accessing the Library E-Reserves Search (https://guides-libraries-psu-edu.ezaccess.libraries.psu.edu/er.php?b=c) and search for your instructor's last name.

Many of the University Libraries resources can be utilized from a distance. Through the Libraries website, you can

- access magazine, journal, and newspaper articles online using library databases;
- borrow materials and have them delivered to your doorstep—or even your desktop;
- get research help via email, chat, or phone using the Ask a Librarian service; and
- much more.

You must have an active Penn State Access Account to take full advantage of the Libraries' resources and service. The Off-Campus Users page (https://libraries.psu.edu/world) has additional information about these free services.

Technical Requirements and Help

Technical Requirements and Help

Operating System	Canvas, Penn State's Learning Management System (LMS), supports most recent versions of Microsoft Windows and Apple Mac operating systems. To determine if your operating system is supported, please review Canvas' computer specifications (https://community.instructure.com/en/kb/articles/662718-what-are-the-browser-and-computer-requirements-for-instructure-products) .
Browser	Canvas supports the last two versions of every major browser release. It is highly recommended that you update to the newest version of whatever browser you are using. Please note that Canvas does not support the use of Internet Explorer. Students and instructors should choose a different browser to use. To determine if your browser is supported, please review the list of Canvas Supported Browsers (https://community.instructure.com/en/kb/articles/662718-what-are-the-browser-and-computer-requirements-for-instructure-products) . --- **Note**: Cookies must be enabled, and pop-up blockers should be configured to permit new windows from Penn State websites.

Additional Canvas Requirements	For a list of software, hardware, and computer settings specifically required by the Canvas LMS, please review Canvas' computer specifications (https://community.canvaslms.com/t5/Canvas-Basics-Guide/What-are-the-browser-and-computer-requirements-for-Instructure/ta-p/66) .
Additional Software	All Penn State students have access to Microsoft Office 365 (https://student.worldcampus.psu.edu/a-z-index/office-365) , including Microsoft Office applications such as Word, Excel, and PowerPoint. Students will need a PDF reader, such as Adobe Reader (https://get.adobe.com/reader/) .
Hardware	**Monitor:** Monitor capable of at least 1024 x 768 resolution **Audio:** Microphone, Speakers **Camera** (optional, recommended)**:** Standard webcam **-** many courses may require a webcam for assignments or exam proctoring software.
Mobile Device (optional)	The Canvas mobile app is available for versions of iOS and Android. To determine if your device is capable of using the Canvas Mobile App, please review the Canvas Mobile App Requirements (https://community.instructure.com/en/kb/articles/388741-unknown) .
Help	If you need technical assistance at any point during the course, please contact the Service Desk (https://www.it.psu.edu/support/) .

Student Education Experience Questionnaire (SEEQ)

During the semester you will receive information about completing the Student Education Experience Questionnaire (SEEQ). Your participation is an opportunity to provide anonymous feedback on your learning experience. Your feedback is important because it allows us to understand your experience in this course and make changes to improve the learning experiences of future students. Please monitor email and course communications for links and availability dates.

Course Requirements and Grading

Descriptions of each element, as well as evaluation criteria, will be provided throughout the course. The dates listed for assignments may change as the semester progresses. Nonetheless, assignments are expected to be completed on time. Final projects turned in more than 24 hours late will be assigned a 10% penalty, and those turned in 1 week or more late will be given a 50% penalty. However, any exceptions to this policy are made at the instructor’s discretion. If you have conflicts or something unexpected arises, do not hesitate to call the instructor to arrive at a solution together. Given the peer interactions required for many of the modules, if you are ill or have a serious problem that prevents your completing an assignment on time, please contact the instructor prior to the class/due date to discuss alternatives.

Students with incomplete assignments at the end of the course will be given the earned final grade. Incomplete or deferred grades (DFs) will not be given except under extenuating circumstances that are discussed with the instructor prior to assignment of final grades.

Any student in this course who has a disability that may prevent them from fully demonstrating their abilities should contact the instructor personally, as soon as possible, to discuss accommodations necessary to ensure full participation and facilitate your educational opportunity.

Course Requirements

Course assignments have been developed to engage class members in critical examination of issues, tools, literature, and design of constructivist learning environments. Combined course assignments are worth 100 points.

Course Requirements

Requirement	Point Value	Module Due (subject to change)
Reading summaries	18 pts (2 pts each)	Module 1, 2, 3, 4, 5, 8, 9, 11, 13
Discussion forums	8 pts (2 pts each)	Module 1, 2, 4, 5
WISE activity	5 pts	Module 3
Design blueprints	60 pts (15 pts each)	Module 7, 10, 12, 14
Peer reviews to blueprints	9 pts (3 pts each)	Module 7, 11, 13
Total	**100 points**	

Weekly Participation

Participation in class is necessary and expected by all class members. Participation will be assessed through a variety of ongoing course activities, including reading summaries, discussions, activity, and peer reviews of blueprints design work.

Design Blueprints

Students will work alone or in small groups to create an instructional design blueprint (i.e., an extended lessons plan connecting theory and design) for selected theoretical models explored during this course. As a class, we will conduct constructive critiques of one another's products and respond reflectively.

Writing Standards

All written work should be of academic quality. Proofread your papers before turning them in. Format the references according to the APA standards (https://guides.libraries.psu.edu/apaquickguide) .

Grading Scale

% of Points	Letter Grade
93–100	A
90–92.9	A−
86–89.9	B+
83–85.9	B
80–82.9	B−
77–79.9	C+
70–76.9	C
60–69.9	D
0-59.9	F

Please refer to the University Registrar's (https://www.registrar.psu.edu/grades/index.cfm) information about University grading policies. If, for reasons beyond the student's control, a student is prevented from completing a course within the prescribed time, the grade in that course may be deferred with the concurrence of the instructor. The symbol DF appears on the student's transcript until the course has been completed. Non-emergency permission for filing a deferred grade must be requested by the student before the beginning of the final examination period. In an emergency situation, an instructor can approve a deferred grade after the final exam period has started. Under emergency conditions during which the instructor is unavailable, authorization is required from one of the following: the dean of the college in which the candidate is enrolled; the executive director of the Division of Undergraduate Studies if the student is enrolled in that division or is a provisional student; or the campus chancellor of the student's associated Penn State campus.

For additional information please refer to the Deferring a Grade (http://student.worldcampus.psu.edu/deferring-grade) page.

Course Schedule

Note: All due dates reflect North American eastern time (ET).

To access the E-Reserves referred to in the Course Schedule, select the Library Resources link in your Course Navigation Menu.

Unit 1: Foundations

Module 0: Exploring Canvas/Getting Started	
Readings	• None

Assignments	• Complete introductions and explore course website (ungraded)

Module 1: Learning Theory (Part I) Constructivist Perspectives

Readings	**Required readings:** • Bransford, J. D., Brown, A. L., & Cocking, R. R. (Eds.). (2000). Learning: From speculation to science. In *How people learn: Brain, mind, experience, and school* (pp. 3–28). National Academy of Sciences. • Bransford, J. D., Brown, A. L., & Cocking, R. R. (Eds.). (2000). Learning and transfer. In *How people learn: Brain, mind, experience, and school* (pp. 51–78). National Academy of Sciences. **Additional resources (recommended):** • Reimann, P., & Markauskaite, L. (2018). Expertise. In F. Fischer, C. Hmelo-Silver, S. Goldman, & P. Reimann (Eds.), *International handbook of the learning sciences* (pp. 54–63). Taylor & Francis. This chapter provides a brief overview of expertise and transfer, and it ends with a description of the process of developing teaching expertise. (**Available in Library E-Reserves**)
Assignments	**Assignments are due Tuesday, 11:59 p.m. (ET)** • Module 1 Summary of Readings • Module 1 Discussion: How People Learn

Module 2: Learning Theory (Part II) Socio-Cultural Perspectives

Readings	**Required reading:** • Barab, S. A., & Duffy, T. M. (2012). From practice fields to communities of practice. In D. H. Jonassen & S. M. Land (Eds.), *Theoretical foundations of learning environments* (2nd ed., pp. 29–65). Routledge. (**Available in Library E-Reserves**) **Choose one from the following supplemental readings, based on relevance to your career/interests:** • Hoadley, C. (2012). What is a community of practice and how can we support it? In D. H. Jonassen & S. M. Land (Eds.), *Theoretical foundations of learning environments* (2nd ed., pp. 286–300). Routledge. (**Available in Library E-Reserves**) One of the most important concepts in social or situated learning theory is the notion of a community of practice. The concept, like the concept of constructivism or the concept of zones of proximal development, has been used both as an explanatory framework for learning and as a metaphor for how instruction should take place. In this chapter, I first describe some basic history of the concept of communities of practice and some of the theoretical assumptions underlying it. Next, I examine some of the key processes identified as producing and sustaining communities of practice. I compare and contrast the notion of communities of practice with other social learning approaches or knowledge communities. And finally, I describe some of the techniques people have used to support communities of practice through technology. • Nasir, N. S., Rosebery, A. S., Warren, B., & Lee, C. D. (2014). Learning as a cultural process: Achieving equity through diversity. In R. K. Sawyer (Ed.), *The Cambridge handbook of the learning sciences* (2nd ed., pp. 686–706). Cambridge University Press. https://doi.org/10.1017/CBO9781139519526.04 (http://ezaccess.libraries.psu.edu/login?url=https://doi.org/10.1017/CBO9781139519526.041) In this chapter, we draw on empirical research on the cultural nature of learning, including studies of (a) learning in and out of school settings; (b) relationships between everyday and academic knowledge and discourse, especially for youth from nondominant groups; and (c) classroom-based design research that explores linkages among the varied repertoires of practice of youth and those of academic disciplines. These studies address multiple dimensions of learning including cognition, discourse, affect, motivation, and identity. Our cultural view of learning is closely related to work on adaptive expertise (Hatano & Inagaki, 1986), that is, the development of flexible knowledge and dispositions that facilitate effective

navigation across varied settings and tasks. Adaptive expertise is crucial for youth from nondominant groups who typically face and must be able to address extreme societal challenges.

- Chinn, C.C., & Selu, G. S. (2019). Bridging the cultural gap in the global workplace. In V. Kenon & S. Palsole (Eds.), *The Wiley handbook of global workplace learning*. Wiley & Sons, Inc.

Over the past three decades, there has been an increase in trade partnerships between countries and an expansion of global markets. As a result of this growth, multinational corporations have faced the challenge of hiring staff with the cultural fluency, sensitivity, and adaptiveness to work at different locations across the globe. Identifying and adapting to workplace culture (situated within ethnic or national culture) and understanding learning in practice are both fundamental focal points within sociocultural theory (Lave & Wenger, 1991; Wertsch, Del Rio, & Alvarez, 1995). In this chapter, we argue that sociocultural theory can provide helpful tools for those who study, create, and manage global workplace learning. We identify some existing challenges in global workplace learning, offer a definition of organizational/workplace culture informed by sociocultural theory, and review some basic assertions of sociocultural theory. Then we identify three features of workplace culture from a sociocultural perspective—training versus learning, artifacts of practice and technology‐facilitated communication—and draw from existing published case studies across diverse contexts to illustrate each of these features.

Assignments	**Assignments due Tuesday, 11:59 p.m. (ET)** • Module 2 Summary of Readings • Module 2 Discussion: Communities of Practice and Practice Fields

Module 3: Technology-Based, Student-Centered Learning Environments

Readings	**Required readings:** • Land, S. M., Hannafin, M. J., & Oliver, K. (2012). Student-centered learning environments: Foundations, assumptions and design. In D. H. Jonassen & S. M. Land (Eds.), *Theoretical foundations of learning environments* (2nd ed., pp. 3–26). Routledge. (**Available in Library E-Reserves**) • Linn, M. C., McElhaney, K. W., Gerard, L., & Matuk, C. (2018). Inquiry learning and opportunities for technology. In F. Fischer, C. E. Hmelo-Silver, S.R. Goldman, & P. Reimann (Eds.), *International handbook of the learning sciences* (pp. 221–233). Routledge. **Additional resources (recommended):** • Lyons, L. (2018). Supporting informal STEM learning with technological exhibits: An ecosystemic approach. In F. Fischer, C. E. Hmelo-Silver, S.R. Goldman, & P. Reimann (Eds.), *International handbook of the learning sciences* (pp. 234–245). Routledge.
Assignments	**Assignments due Tuesday, 11:59 p.m. (ET)** • Module 3 Summary of Readings • Module 3 Discussion: Web-Based Inquiry Science Environment (WISE) Project Examples

Module 4: Constructionism, Technology, and Learning-by-Constructing

Readings	**Required readings:** • Berland, M., Halverson, E., Wilkerson, M., & Polman, J. (2017). Expressive construction: Enabling learners to represent powerful ideas. In J. Roschelle, W. Martin, J. Ahn, & P. Schank (Eds.), *Cyberlearning community report: The state of cyberlearning and the future of learning with technology.* SRI International. • Halverson, E., & Peppler, K. (2018). The maker movement and learning. *International Handbook of the Learning Sciences.* (**Available in Library E-Reserves**) **Additional resources (recommended):** • Halverson, E., & Sheridan, K. (2014). The maker movement in education. *Harvard Educational Review, 84*(4), 495–504. In this essay, Erica Halverson and Kimberly Sheridan provide the context for research on the maker movement as they consider the emerging role of making in education. The authors describe the theoretical roots of the movement and draw connections to related research on formal and informal education. They present points of tension between making and formal education practices as they come into contact with one another, exploring whether the newness attributed to the maker movement is really all that new and reflecting on its potential pedagogical impacts on teaching and learning. • Ackermann, E. (2001). Piaget's constructivism, Papert's constructionism: What's the difference? www.sylviastipich.com/wp-content/uploads/2015/04/Coursera-Piaget-_-Papert.pdf What is the difference between Piaget's constructivism and Papert's "constructionism"? Beyond the mere play on the words, I think the distinction holds, and that integrating both views can enrich our understanding of how people learn and grow. Piaget's constructivism offers a window into what children are interested in, and able to achieve, at different stages of their development. The theory describes how children's ways of doing and thinking evolve over time, and under which circumstance children are more likely to let go of—or hold onto—their currently held views. Piaget suggests that children have very good reasons not to abandon their worldviews just because someone else, be it an expert, tells them they're wrong. Papert's constructionism, in contrast, focuses more on the art of learning, or "learning to learn," and on the significance of making things in learning. Papert is interested in how learners engage in a conversation with (their own or other people's) artifacts, and how these conversations boost self-directed learning, and ultimately facilitate the construction of new knowledge. He stresses

	the importance of tools, media, and context in human development. Integrating both perspectives illuminates the processes by which individuals come to make sense of their experience, gradually optimizing their interactions with the world. • Bar-El, D., & Worsley, M. (2019, June). *Tinkering with Music: Designing a maker curriculum for an after-school youth club*. IDC '19: Proceedings of the 18th ACM International Conference on Interaction Design and Children, Boise, ID, United States. https://doi.org/10.1145/3311927.3323127 (**Available in Library E-Reserves**) In recent years, researchers have focused on the design and implementation of maker activities across formal and informal settings. As a result, the research community is gradually articulating the challenges and design considerations relating to these settings. These include: tools, facilitation, and curricular requirements. In this paper we present the design and implementation of Tinkering with Music, a 10-week youth club curriculum around popular music appreciation and instrument building with electronics. Reflecting on our design and implementation, we report on: (1) our curricular activities; (2) design challenges which we had to overcome throughout implementation; and (3) a failure to engender long-term engagement with tools and practices of the curriculum.
Assignments	**Assignments due Tuesday, 11:59 p.m. (ET)** • Module 4 Summary of Readings • Module 4 Discussion: Learning by Constructing

Unit 2: Blueprint Design Activities

Module 5: Teaching for Understanding (TFU) Design Framework

Readings	**Required reading:** • Perkins, D., & Unger, C. (1999). Teaching and learning for understanding. In C. M. Reigeluth (Ed.), *Instructional-design theories and models: A new paradigm of instructional theory* (3rd ed., pp. 91–114). (**Available in Library E-Reserves**) You may also read more about the project on the Teaching for Understanding page. **Additional resources (recommended):** • Perkins, D. (1993). Teaching for understanding. *American Educator: The Professional Journal of the American Federation of Teachers*, *17*(3), 8, 28–35.
Assignments	**Assignments due Tuesday, 11:59 p.m. (ET)** • Module 5 Summary of Readings • Module 5 Discussion: Sharing Perspectives on Understanding

Module 6: TFU Design Blueprint/Peer Reviews of Design Blueprints

Readings	None
Assignments	**Assignment due Friday, 11:59 p.m. (ET)** • Post TFU design blueprint to discussion forum **Assignment due Tuesday, 11:59 p.m. (ET)** • Provide peer reviews of classmates' TFU design blueprints

Module 7: Problem-Based Learning

Readings	**Required reading:** • Lu, J., Bridges, S., & Hmelo-Silver, C. E. (2014). Problem-based learning. In R. K. Sawyer (Ed.), *The Cambridge handbook of the learning sciences* (2nd ed., pp. 298–318). Cambridge University Press. https://doi.org/10.1017/CBO9781139519526.019 **Choose one from the following supplemental readings, based on relevance to your career/interests:** • Navy, S., Edmondson, E., Maeng, J., Gongzi, A., & Mannarino, A. (2019). How to create problem-based learning units: Understanding the PBL planning process. *Science and Children.* 56(5), 68-72. This article provides a PBL example and template for creating PBL lessons that are tied to standards. The example presented is in the domain of earth sciences. • O'Brian, E., Hamburg, I., & Southern, M. (2019). Using technology‐oriented, problem‐based learning to support global workplace learning. In V. Kenon & S. Vasant Palsole (Eds.), *The Wiley handbook of global workplace learning* (pp. 591–609). This chapter presents a research paper on PBL in the workplace environment. It discusses the PBL platform and steps for creating PBL, ending with research findings. • Brush, T., & Saye, J. (2008). The effects of multimedia-supported problem-based inquiry on student engagement, empathy, and assumptions about history. *Interdisciplinary Journal of Problem-Based Learning, 2*(1), 21–56. This research investigates how multimedia resources can be designed to support problem-based learning in secondary history classrooms. The focus of the PBL is on the process of learning historical empathy. **Additional resources (recommended):** • Cindy Hmelo-Silver: Problem-Based Learning webinar (1:49:00 in length)

Assignments	**Assignment due Tuesday, 11:59 p.m. (ET)** • Module 7 Summary of Readings • Prepare for the PBL design blueprint

Module 8: PBL as Project-Based Learning

Readings	**Required readings:** • Krajcik, J. S., & Shin, N. (2014). Project-based learning. In R. K. Sawyer (Ed.), *The Cambridge handbook of the learning sciences* (2nd ed., pp. 275–297). Cambridge University Press. https://doi.org/10.1017/CBO9781139519526.018 This chapter presents features of project-based learning environments, based on learning sciences theory. The authors state: "project-based learning allows students to learn by doing, to apply ideas, and to solve problems." The authors discuss examples of their own research in the domain of project-based science. • Schwartz, D., Brophy, S., Lin, X., & Bransford, J. (1999). Software for managing complex learning: Examples from an educational psychology course. *Educational Technology Research & Development, 47*(2), 39–59. https://doi.org/doi.org/10.1007/BF02299464
Assignments	**Assignment due Tuesday, 11:59 p.m. (ET)** • Module 8 Summary of Readings • Prepare for the PBL design blueprint

Module 9: Work on PBL Design Blueprint

Readings	None
Assignments	**Assignments due Tuesday, 11:59 p.m. (ET)** • Post PBL design blueprint to discussion forum

Module 10: Seamless, Mobile Learning

Readings	**Required reading:** • Sharples, M., & Pea, R. (2014). Mobile learning. In R. K. Sawyer (Ed.), *The Cambridge handbook of the learning sciences* (2nd ed., pp. 501–521). Cambridge University Press. https://doi.org/10.1017/CBO9781139519526.030 **Choose one from the following supplemental readings, based on relevance to your career/interests:** • Emerson, L., & Berge, Z. (2018). Microlearning: Knowledge management applications and competency-based training in the workplace. *Knowledge Management & E-Learning, 10*(2), 125–132. The focus of this article is a threefold discussion on microlearning: 1) how microlearning best practices facilitate knowledge acquisition in the workplace by engaging and motivating employees through short, personalized, just-in-time learning; 2) ways microlearning integrates with knowledge management applications through situational mentoring; and 3) how competency-based microlearning, via subscription learning, is both an innovative approach to e-learning and an asset to learning organizations focused on improving the performance of their employees. • Squire, K. (2012). Mobile media learning: Ubiquitous computing environments for the mobile generation. In C. Mouza & N. Lavigne (Eds.), *Emerging technologies for the classroom.* Springer. https://doi.org/doi.org/10.1007/978-1-4614-4696-5_13 The focus of this chapter is on how mobile technologies create new opportunities for learning outside of formal classrooms. Squire presents three case study “exemplars” of using mobile media for learning: (1) an augmented reality game, Saving Lake Wingra, a place-based augmented reality (AR) curriculum unit designed around Lake Wingra in Madison, Wisconsin; (2) an AR game called Mentira, which is used in a Spanish class at University of New Mexico and involves students going out into Spanish-speaking neighborhoods to practice language and better understand culture within communities; and (3) Mobile-Design Workshop, which discusses how high school students create mobile AR gaming experiences. • van’t Hooft, M. (2012). The potential of mobile technologies to connect teaching and learning inside and outside of the classroom. In C. Mouza & N. Lavigne (Eds.), *Emerging technologies for the classroom.* Springer. https://doi.org/doi.org/10.1007/978-1-4614-4696-5_12

This chapter (like Squire above) focuses on how mobile technologies create new opportunities for learning outside of formal classrooms. Three case study "exemplars" of using mobile media for learning are presented: (1) Frequency 1550, a GPS-enabled learning game that allows high school students to learn about the history of Medieval Amsterdam; (2) MyArtSpace, a mobile learning experience that connects classrooms and an Art Museum; and (3) the Geohistorian Project, which involves digital storytelling to help students think like historians.

- Dunleavy, M. (2014). Design principles for augmented reality learning. *TechTrends, 58*(1), 28–34. https://doi.org/10.1007/s11528-013-0717-2

Augmented reality is an emerging technology that utilizes mobile, context-aware devices (e.g., smartphones, tablets) that enable participants to interact with digital information embedded within the physical environment. This overview of design principles focuses on specific strategies that instructional designers can use to develop AR learning experiences. A review of the literature reveals the following three design principles as instructive: (a) enable and then challenge (challenge); (b) drive-by gamified story (fantasy); and (c) see the unseen (curiosity). These design principles can also be viewed as an attempt to either leverage the unique affordances of AR or minimize the limitations of the medium as reported in the literature (Dunleavy & Dede, 2014). As the field matures and more research teams explore the potential of AR to enhance teaching and learning, it will be critical to determine the design techniques that optimize the unique affordances of AR, minimize the limitations of the medium, and ultimately enhance learning across the curriculum.

- Lee, V., & Drake, J. (2013). Quantified recess: Design of an activity for elementary students involving analyses of their own movement data. In J. P. Hourcade, E. A. Miller & A. Egeland (Eds.), *Proceedings of the 12th International Conference on Interaction Design and Children* (pp. 273-276). ACM.

Recess is often a time for children in school to engage recreationally in physically demanding and highly interactive activities with their peers. This paper describes a design effort to encourage fifth-grade students to examine sensitivities associated with different measures of center by having them analyze activities during recess . . . over the course of a week using Fitbit activity trackers and TinkerPlots data visualization software. We describe the activity structure [of] some observed student behaviors during the activity. We also provide a descriptive account, based

	on video records and transcripts, of two students who engaged thoughtfully with their recess data and developed a more sophisticated understanding of when and how outliers affect means and medians. • Zimmerman, H. T., & Land, S. M. (2014). Facilitating place-based learning in outdoor informal environments with mobile computers. *TechTrends, 58*(1), 77–83. This paper advocates for place-based education to guide research and design for mobile computers used in outdoor informal environments (e.g., backyards, nature centers and parks). By bringing together research on place-based education with research on location awareness, we developed three design guidelines to support learners to develop robust science-related understandings within local communities. The three empirically-derived design guidelines are: (1) Facilitate participation in disciplinary conversations and practices within personally-relevant places, (2) Amplifying observations to see the disciplinary-relevant aspects of a place, and (3) Extending experiences through exploring new perspectives, representations, conversations, or knowledge artifacts. Last, we link theory to practice by illustrating how the three guidelines were applied in one outdoor science learning project called Tree Investigators.
Assignments	**Assignment due Friday, 11:59 p.m. (ET)** • Provide peer reviews of classmates' PBL design blueprints **Assignment due Tuesday, 11:59 p.m. (ET)** • Module 10 Summary of Readings • Prepare for the Seamless, Mobile Learning design blueprint

Module 11: Work on Seamless, Mobile Learning Design Blueprint

Readings	None
Assignments	**Assignment due Tuesday, 11:59 p.m. (ET)** • Post Seamless, Mobile Learning design blueprint

Module 12: Learning Communities

Readings	**Required readings:** • Slotta, J. D., & Najafi, H. (2013). Supporting collaborative knowledge construction with Web 2.0 technologies. In C. Mouza & N. Lavigne (Eds.), *Emerging technologies for the classroom: Explorations in the learning sciences, instructional systems and performance technologies* (pp. 93–112). Springer. https://doi.org/10.1007/978-1-4614-4696-5_7 • Greenhow, C., & Li, J. (2013). Like, comment, share: Collaboration and civic engagement within social network sites. In C. Mouza & N. Lavigne (Eds.), *Emerging technologies for the classroom: Explorations in the learning sciences, instructional systems and performance technologies* (pp. 127–141). Springer. https://doi.org/10.1007/978-1-4614-4696-5_9
Assignments	**Assignment due Friday, November 20, 11:59 p.m. (ET)** • Provide peer reviews of classmates' Seamless, Mobile Learning design blueprints. **Assignment due Tuesday, 11:59 p.m. (ET)** • Module 12 Summary of Readings • Prepare for the Learning Communities design blueprint

Module 13: Work on Learning Communities Design Blueprint

Readings	None
Assignments	**Assignment due Tuesday, 11:59 p.m. (ET)** • Post Learning Communities design blueprint to discussion forum

Module 14: Final Reflection

Readings	None
Assignments	**Assignments due Friday, 11:59 p.m. (ET)** • Reflection assignment

Note: If you are planning to graduate this semester, please communicate your intent to graduate to your instructor. This will alert your instructor to the need to submit your final grade in time to meet the published graduation deadlines. For more information about graduation policies and deadlines, please refer to Graduation (https://student.worldcampus.psu.edu/graduating-students/graduation) at the Chaiken Center for Student Success.

Academic Integrity

According to Penn State policy G-9: Academic Integrity (https://aappm.psu.edu/policy/g-9-academic-integrity) (for undergraduate students in undergraduate courses) and policy GCAC-805 Academic Integrity (https://gradschool.psu.edu/graduate-education-policies/gcac/gcac-800/gcac-805-academic-integrity) (for graduate students and undergraduate students in graduate courses), an academic integrity violation is “an intentional, unintentional, or attempted violation of course or assessment policies to gain an academic advantage or to advantage or disadvantage another student academically.” Unless your instructor tells you otherwise, you must complete all course work entirely on your own, using only sources that have been permitted by your instructor, and you may not assist other students with papers, quizzes, exams, or other assessments. If your instructor allows you to use ideas, images, or word phrases created by another person (e.g., from Course Hero or Chegg) or by generative technology, such as ChatGPT, you must identify their source. You may not submit false or fabricated information, use the same academic work for credit in multiple courses, or share instructional content. Students with questions about academic integrity should ask their instructor **before submitting work.**

Students facing allegations of academic misconduct may not drop/withdraw from the affected course unless they are cleared of wrongdoing (see G-9: Academic Integrity (https://aappm.psu.edu/policy/g-9-academic-integrity) or GCAC-805 Academic Integrity (https://gradschool.psu.edu/graduate-education-policies/gcac/gcac-800/gcac-805-academic-integrity) as appropriate). Attempted drops will be prevented or reversed, and students will be expected to complete course work and meet course deadlines. Students who are found responsible for academic integrity violations face academic outcomes, which can be severe, and put themselves at jeopardy for other outcomes which may include ineligibility for Dean’s List, pass/fail elections, and grade forgiveness. Students may also face consequences from their home/major program and/or The Schreyer Honors College.

How Academic Integrity Violations Are Handled
World Campus students are expected to act with civility and personal integrity; respect other students' dignity, rights, and property; and help create and maintain an environment in which all can succeed through the fruits of their own efforts. An environment of academic integrity is requisite to respect for oneself and others, as well as a civil community.

In cases where academic integrity is questioned, procedures allow a student to accept or contest/appeal the allegation. If a student chooses to contest/appeal the allegation, the case will then be managed by the respective school, college or campus Academic Integrity Committee. Review procedures may vary by college, campus, or school, but all follow the aforementioned policies.

All academic integrity violations are referred to the Office of Student Accountability and Conflict Response (https://studentaffairs.psu.edu/student-accountability) , which may assign an educational intervention and/or apply a Formal Warning, Conduct Probation, Suspension, or Expulsion.

Information about Penn State's academic integrity policy is included in the information that students receive upon enrolling in a course. To obtain that information in advance of enrolling in a course, please contact us by going to the Contacts & Help page (http://student.worldcampus.psu.edu/contacts-help-current-students) .

Student Success and Support Resources

The Chaiken Center for Student Success at Penn State World Campus (https://student.worldcampus.psu.edu/) guides you to the right resources and support you need–when you need them–along your academic journey. You can connect with peers and support teams to find direction, information, and networking opportunities. On the website, you'll find information and resources on many aspects of being a World Campus student:

- **Finances**—tuition, scholarships, and financial aid
- **Inclusion and Wellness**—diversity and inclusion, mental health services, disability accommodations, care and advocacy
- **Enrollment and Registration—**course planning, adding and dropping courses, and much more
- **Course Work and Success—**academic advising, tutoring, and other services
- **Involvement and Opportunities—**career resources, student organizations, internships, service, study abroad, and more

Following are some key resources.

Student Disability Services

Penn State welcomes students with disabilities, whether permanent or temporary, visible or hidden, into the University's educational programs. Every Penn State campus has an office for students with disabilities, including World Campus. The Disabilities and Accommodations (https://student.worldcampus.psu.edu/inclusion-and-wellness/disabilities-and-accommodations) section of the Chaiken Center for Student Success website provides World Campus students with information regarding how to request accommodations, documentation guidelines and eligibility, and appeals and complaints. For additional information, please visit the University's Student Disability Resources website (http://equity.psu.edu/sdr) .

In order to receive consideration for reasonable accommodations, you must contact the appropriate disability services office (https://equity.psu.edu/offices/student-disability-resources/campus-offices) at the campus where you are officially enrolled, participate in an intake interview, and provide documentation. If the documentation supports your request for reasonable accommodations, your campus's disability services office will provide you with an accommodation letter. Please share this letter with your instructors and discuss the accommodations with them as early in your courses as possible. You must follow this process for every semester that you request accommodations.

Graduate students: Please see the Student Disability Resources website for graduate student resources (https://equity.psu.edu/offices/student-disability-resources/graduate-students-with-disabilities) .

Students with disabilities participating in internship, practicum, student teaching, or other experiential learning opportunities as part of their degree requirements may also be eligible for reasonable accommodations to ensure equal access and opportunity. These accommodations are determined through an interactive process involving the student, their University supervisor, and the site supervisor. Student Disability Resources can assist students with identifying potential barriers, facilitating accommodation requests, and coordinating with University supervisors to promote inclusive learning experiences.

Counseling and Psychological Services

If you have a crisis or safety concern, mental health services (http://student.worldcampus.psu.edu/inclusion-and-wellness/mental-health) are available to you as a Penn State student. Crisis and emergency contacts (https://student.worldcampus.psu.edu/help-and-support/crisis-and-emergency-help) are available, no matter where you are located:

- **Anywhere in the United States:** Call the Penn State Crisis Line at 1-877-229-6400 or text LIONS to 741741. You can also contact your local crisis services or hospital for emergencies.
- **Outside the United States**: Please contact emergency services in your current location. You can also use the International Crisis and Emergency Services listings (https://findahelpline.com/) .
- **At University Park:** Assistance is available at Counseling & Psychological Services (CAPS) locations (https://studentaffairs.psu.edu/counseling/UP-locations) on campus.
- **At a Penn State branch campus:** You can search for counseling information at your campus (https://studentaffairs.psu.edu/counseling/caps-campuses) .

Military Student Information

Veterans and currently serving military personnel and/or dependents with unique circumstances (e.g., upcoming deployments, drill/duty requirements, VA appointments, etc.) are welcome and encouraged to communicate these, in advance if possible, to the instructor in the case that special arrangements need to be made.

Course Policies

Privacy Notice

In order to protect your privacy, course access is limited to those individuals who have direct responsibility for the quality of your educational experience. In addition to the instructor, a teaching assistant or college administrator may be provided access in order to ensure optimal faculty availability and access. World Campus technical staff may also be given access in order to resolve technical support issues.

Student Responsibilities and Conduct

1. Students are responsible for online course content, taking notes, obtaining other materials provided by the instructor, taking tests (if applicable), and completing assignments as scheduled by the instructor. As a general rule, students should plan on logging into the course at least three times per week and spending at least three hours per course credit per week on the course, e.g., if the course is three credits, the student should plan on spending at least 9-12 hours per week on the course, just as they would in a residence course.
2. Students are responsible for keeping track of changes in the course syllabus made by the instructor throughout the semester.
3. Students are responsible for monitoring their grades.
4. Students must contact their instructor (and teammates when working on any collaborative learning assignments) as soon as possible if they anticipate missing long periods of online time due to events such as chronic illnesses, death in the family, business travel, or other appropriate events. The instructor will determine the minimal log on time and participation required in order to meet course responsibilities. In the event of other unforeseen conflicts, the instructor and student will arrive at a solution together.
    a. Instructors may require students to provide documentation with the class absence form or other written notification for events such as illness, family emergency, or a business-sanctioned activity.
    b. Conflicts with dates on which examinations or assignments are scheduled must be discussed with the instructor or TA (teaching assistant) prior to the date of the examination or assignment.

5. Students are responsible for following appropriate netiquette (network etiquette) when communicating with their instructor and classmates. For reference, see the Academic Success Kit (https://student.worldcampus.psu.edu/course-work-and-success/academic-skills) .
6. Behaviors that disrupt other students' learning are not acceptable and will be addressed by the instructor.
7. For severe and chronic problems with student disruptive behavior, the following will be applied for resolution:
    a. Senate Committee on Student Life policy on managing classroom disruptions: Office of Student Accountability and Conflict Response. (https://studentaffairs.psu.edu/student-accountability)
    b. Penn State Values. (https://universityethics.psu.edu/penn-state-values)

Report Bias

Penn State takes great pride to foster a diverse and inclusive environment for students, faculty, and staff. Acts of intolerance, discrimination, or harassment due to age, ancestry, color, disability, gender, gender identity, national origin, race, religious belief, sexual orientation, or veteran status are not tolerated and can be reported through Educational Equity via the Report Bias webpage (http://equity.psu.edu/reportbias/) .

Disclaimer: Please note that the specifics of this Course Syllabus are subject to change, and you will be responsible for abiding by any such changes. Your instructor will notify you of any changes.

Please direct questions to the IT Service Desk (https://www.it.psu.edu/support/) | The Pennsylvania State University © 2026